

Global Facilities - Design & Construction Standard - Gas - UHP Stainless Steel for Bulk and Specialty Gases

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1.0 Purpose

- 1.1 This standard defines the minimum material specifications, manufacturing, cleaning, packaging, and installation requirements for ultrahigh purity (UHP) stainless steel materials and components utilized to supply and distribute bulk and specialty gases in the semiconductor manufacturing process.
- 1.2 This document applies to all materials and components utilized to supply and distribute fluids to the manufacturing process by stainless steel systems. These fluids include bulk gases, bulk and cylinder specialty gases, organic liquid chemicals, and precursors based on, but not limited to, the following: TEOS, TEPO, TEB, TiCl₄, TMA, TDMAH, TEMZ, Pyridine, Hexachloro-Disilane (HCDS) and others.
- 1.3 This document is intended to supplement the Construction Drawings and Specifications for construction projects and purchase specifications for the supply and distribution of bulk and specialty gases.
- 1.4 All deviations from this standard shall be documented by the site team including justification and shall be reviewed and approved by Corporate GFTT and FCT. Refer to **Appendix 1 – Template Form for Site Specific Deviation List**.

- 1.5 For sites that require higher standards, specifications, or regulations for compliance, etc., those sites may have to follow their existing (or better) standards, specifications, or regulations to meet the site-specific requirements. Any deviations from this standard due to site specific requirements shall be documented in the "Site Specific Deviation List" in **Appendix 1**.
- a) During the design development stage of a new, greenfield construction project, the vendor, contractor, and/or project team shall review the Global System Standard and create the list of proposed deviations.
 - b) Project team shall review and perform a risk assessment for each deviation using RPN SOD criteria, then obtain approval from Project Manager.
 - c) The Deviation List shall be reviewed and approved by all stakeholders (GFTT, FCT, OCT, EHS, FAB Support, etc.) prior to Global Facilities (GFTT & FCT) Directors' review.
 - d) After approval from Global Facilities Directors, the deviation list shall be reviewed and approved by Global Construction and Site Facilities Directors before authorization to proceed.

2.0 Scope

- 2.1 This document provides for the design and specification of the purchase and installation of UHP Stainless Steel materials and components used in the supply and distribution of bulk and specialty gases for new Greenfield construction projects.
- 2.2 For improvements or upgrades to existing gas systems, consult FCT for Specifications to follow.
- 2.3 This document applies to all Micron Facilities employees, contractors, and vendors at all Micron sites for new Greenfield FAB Construction Projects. Based upon requirements and applicability with existing design, this document may also be used for Brownfield or other projects upon review with FCT.

3.0 Definitions and Acronyms

Term/Acronym	Definition / Description
FCT	Facilities Central Team. This is used interchangeably with Global Facilities Engineering Team and associated to O&M BKM's
GFTT	Global Facilities Technology Team. This is used interchangeably with Global Facilities Engineering Team and associated to design standards
GFC	Global Facilities & Construction Team. This is used interchangeably with Global Facilities (GFac), Facilities Central Group (FCG), World Wide Operations (WWOps), World Wide Facilities (WW_FACILITIES), Corp Facilities
Global Facilities Central Team	This is used interchangeably with Innovation & Integration Team
Global Facilities Construction Team	This is used interchangeably with Global Construction Team
Global Facilities Operations Support Team	This is used interchangeably with Global Facilities Operations Team
Global Facilities Planning Team	This is used interchangeably with Global Facilities Line Planning Team
BA	Bright Annealed
BIC	Building Isolation Cabinet
BSGS	Bulk Specialty Gas Supply
DVB / DVMB	Distribution Valve Box / Distribution Valve Manifold Box
EHS	Environmental Health and Safety
EP	Electropolished
EPCM	Engineering, Procurement, Construction Management
FCT	Facilities Central Team
GFTT	Global Facilities Technology Team
GIB	Gas Isolation Box
GN2	General Nitrogen
GO2	General Oxygen
GTAW	Gas Tungsten Arc Weld
HPM	Hazardous Production Material
ISO	International Standards Organization
OCT	Operations Central Team

Term/Acronym	Definition / Description
PAr	Process Argon
PCTFE	Polychlorotrifluoroethylene (commonly referred to as Kel-F® or Neoflon®)
PH2	Process Hydrogen
PI	Polyimide (commonly referred to as Vespel®)
PN2	Process Nitrogen
PO2	Process Oxygen
PPE	Personal Protective Equipment
PTFE	Polytetrafluoroethylene (commonly referred to as Teflon)
RA	Average Roughness
UHP	Ultra-High Purity
VCR	Vacuum Coupling Radiation (a type of metal gasket face seal fitting)
VIM	Vacuum Induction Melting (Single Melt)
VIM-VAR	Vacuum Induction Melting & Vacuum Arc Remelting (Double Melt)
VMB	Valve Manifold Box

4.0 Roles and Responsibilities

Roles	Responsibilities
GFTT Lead	Develop and maintain System Standard document • Incorporate Best Known Methods (BKM), lessons learned, and emerging technologies in the System Standard • Perform periodic document review and update System Standard • Support the Site teams by providing knowledge and technical expertise • Review site specific deviations from this system standard
FCT Lead	Review and provide technical feedback on System Standard Update BKMs and Develop relevant BKMs • Support the Site teams by providing knowledge and technical expertise in operation and maintenance • Review site specific deviations from this system standard
Site Project Construction Team	• Review and provide technical feedback on System Standard • Incorporate System Standard into Project Engineering, Design, and Construction • Document project deviations, exceptions, and exclusions from System Standard • Share new lessons learned to GFTT Lead, to determine if it is Best Known Method (BKM) that can be captured in future revision of System Standard • Consult with Global/Site PSM on the Relative Ranking Analysis (RRA) and ensure the RRA is less than 30. Otherwise carry out mitigation till RRA comes below 30.
Site Facilities Team	• Use System Standard as technical reference for new system designs • Identify and document deviations from the system standard/specification for review and approval by GFTT and FCT.
Equipment Supplier or other contractors	• Meet or exceed the requirements as documented in Micron standards • Identify and justify non-compliance or deviations from system standard for which supplier or contractor is unable to meet, and recommend alternate solution for review by site team, GFTT, and FCT.
Global/Site PSM	• Provide guidance for Relative Ranking Analysis rating for systems • Consult on the required mitigation

5.0 Project Safety Requirements

- 5.1 The EPCM contractor or system vendor's written EHS program shall contain all necessary elements to administer the program in accordance with all applicable EHS Requirements, Micron Corporate and Site-specific safety policies as well as local regulatory requirements. Compliance with the below mentioned link to Micron Safety Standards shall also apply.
- [Global EHS – Construction Contractor EHS Minimum Performance Requirements](#)
 - [Global EHS – Construction EHS Management Strategy](#)
 - [Global EHS – Contractor EHS Requirements](#)
 - [Global EHS – Lifting and Rigging Standard](#)
 - [Global EHS – Confined Space Program Standard](#)
 - [Global EHS – Control of Hazardous Energy \(CoHE\) Standard](#)
 - [Global EHS – Electrical Safety Standard](#)
 - [Global EHS – Work at Heights Standard](#)
 - [Global EHS – Lone Worker Standard](#)
- 5.2 Contractors/system vendors and Micron site project team are expected to assess all related safety risk factors during startup and commissioning activities. Preventive and contingency plans shall be created to address all safety concerns. Job Hazard Analysis (JHA) or Risk Assessments shall be provided as necessary.
- 5.3 All contractor's and system vendor's personnel shall adhere to their own mandated PPE requirements for HPM gases and shall be reviewed with Micron site's Environmental Health and Safety (EHS) for adequacy.
- 5.4 Examples of major safety concerns include, but are not limited to:
- General area safety (e.g., tripping hazards)
 - Potential HPM exposure / leaks
- 5.5 Contractors/system vendors are expected to ensure that their team members and sub-contractors are adequately trained in handling HPM materials. This should include all necessary in-house and specialized training specific to equipment or system.

6.0 Codes and Standards

Code/Standard	Title
ASTM A269	Standard Specification for Seamless and Welded Austenitic Stainless Steel Tubing for General Service
ASTM A312	Standard Specification for Seamless, Welded, and Heavily Cold Worked Austenitic Stainless Steel Pipes
ASTM A632	Standard Specification for Seamless and Welded Austenitic Stainless Steel Tubing (Small-Diameter) for General Service
ASTM G93	Standard Practice for Cleaning Methods and Cleanliness Levels for Material and Equipment Used in Oxygen-Enriched Environments
JIS G3463	Japanese Industrial Standard – Stainless Steel Tubes
JIS G3459	Japanese Industrial Standard – Stainless Steel Pipes
AWS D10.9	Specification and Qualification of Welding Procedures and Welders for Piping and Tubing
ANSI/ASME B31.3	Chemical Plant and Petroleum Refining Piping, Boiler and Pressure Vessel Code – Section IX, Qualification Standard for Welding and Brazing Procedures, Welders, Brazers, and Welding and Brazing Operators
ANSI/AWS A5.32	Specification for Welding Shielding Gases
AIGA 021/12	Oxygen Pipeline and Piping Systems
AIGA 087/14	Standard for Hydrogen Piping Systems at User Locations
AWS D10.9	Specification and Qualification of Welding Procedures and Welders for Piping and Tubing
SEMI F20	Specification for 316L Stainless Steel Bar, Forgings, Extruded Shapes, Plate, and Tubing for Components Used in General Purpose, High Purity and Ultra-High Purity Semiconductor Manufacturing Applications
2W4373RQWREN-1568922467-11	Global EHS – Toxic Gas Monitoring and Double Containment Standard

7.0 Material and Component Specifications

- 7.1 Stainless steel tubing shall conform to the specifications of ASTM A269 for tube sizes $\geq \frac{1}{2}$ inch, and ASTM A632 for tube sizes $< \frac{1}{2}$ inch, or JIS G3463.
- 7.2 Stainless steel pipe shall conform to the specifications of ASTM A312 or JIS G3459.
- 7.3 All pipe and tube for specialty gases and process bulk gases (purified) shall be produced from ASTM TP316L or JIS SUS316LTP grade material as a minimum.
- 7.4 All pipe and tube for general and utility grade bulk gases (pre-purification or unpurified) shall be produced from ASTM 304L or JIS SUS304LTP grade material as a minimum.
- 7.5 Chemical Composition for Stainless Steel Material shall comply with ASTM and JIS specifications as below:

Spec	Grade	Chemical Composition %							
		C	Si	Mn	P	S	Ni	Cr	Mo
A269, A312, A632	TP 304L	0.035 max (1)	1.00 max	2.00 max	0.045 max	0.030 max	8.00- 12.00	18.00- 20.00	--
G3459, G3463	SUS304LTP SUS304LTB	0.030 max	1.00 max	2.00 max	0.045 max	0.030 max	9.00- 13.00	18.00- 20.00	--
A269, A312, A632	TP 316L	0.035 max (1)	1.00 max	2.00 max	0.045 max	0.030 max	10.00- 15.00	16.00- 18.00	2.00- 3.00
G3459, G3463	SUS316LTP, SUS316LTB	0.030 max	1.00 max	2.00 max	0.045 max	0.030 max	12.00- 16.00	16.00- 18.00	2.00- 3.00

Note 1: For small diameter or thin walls, a carbon maximum of 0.040% is necessary in grades TP304L and TP316L. Small diameter tubes are defined as those with less than 0.5 inch in outside diameter and thin walls are those less than 0.049 inch in minimum wall thickness.

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- 7.6 Recommend the sulfur content of 316L SS single melt (VIM) and double melt (VAR) tubing and fittings is within 0.005% - 0.017%.
- 7.7 All UHP stainless steel tube for process service gases (i.e., PN₂, PO₂, PH₂, PAr, specialty gases, etc.) shall be manufactured from 316L stainless steel, single melt material and electropolished with an interior surface finish of no greater than 20 µin (0.50 µm) RA max and 10 µin (0.25 µm) RA average at a minimum.
- 7.8 All UHP stainless steel tube or pipe for general service gases (i.e., GN₂, GO₂, etc.) or secondary containment shall be manufactured from 304L stainless steel material as a minimum.
- 7.9 For tube sizes ≤ 1 inch, seamless tubing shall be used.
- 7.10 For tube sizes > 1 inch and ≤ 2 inches, tubing may be seamless or welded according to manufacturer's capability.
- 7.11 For tube sizes > 2 inches, welded tubing shall be used.
- 7.12 Tubing used in general service (i.e., GN₂, GO₂, GAr, etc.) shall have a bright annealed interior surface finish with 40 µin (1 µm) RA average as a minimum. Additionally, tubing used in any oxygen service shall be cleaned for oxygen service according to CGA G-4.1.
- 7.13 Electropolished tubing (single melt and double melt) shall have an interior surface finish of 10 µin (0.25 µm) average RA, with no single measurement greater than 20 µin (0.50 µm) RA maximum.
- 7.14 For use in non-HPM process, inert and asphyxiant gases, all UHP stainless steel components (i.e., valves, regulators, fittings, etc.) shall be manufactured from 316L stainless steel single melt (VIM) material as a minimum, and electropolished to 10 µin (0.25 µm) Ra average, and 20 µin (0.50 µm) Ra maximum internal finish.
- 7.15 For use in toxic, corrosive, or oxidizer HPM process gas services, recommend all UHP stainless steel components (i.e., valves, regulators, fittings, etc.) to be manufactured using 316L stainless steel double melt (VIM-VAR) material, and electropolished to 10 µin (0.25 µm) Ra average, and 20 µin (0.50 µm) Ra maximum internal finish for superior corrosion resistance. At a minimum, 316L stainless steel single melt material, electropolished to 10 µin (0.25 µm) Ra average, and 20 µin (0.50 µm) Ra maximum internal finish is allowed.
- 7.16 Additional requirements for organic solvents dispense and distribution systems (i.e., IPA, PGMEA, HMDS, etc.)
- a) Chromium content in the 316L stainless steel shall be between 16% - 18%.
 - b) Sulfur content in seamless tubing shall be between 0.005% - 0.012%.
 - c) Sulfur content in welded tubing shall be between 0.005% - 0.017%.

- d) Butt weld fittings and components may have sulfur content lower than 0.005% according to manufacturer's standards; however, the welding program must be re-validated, and coupons must be welded and inspected according to the QC acceptance criteria before installing these micro fittings in UHP stainless steel systems.
- e) Mechanical compression fittings are allowed for organic solvent dispense and distribution systems, such as IPA, PGMEA systems, and others based on engineering standard, site construction specifications, and directed in the design phase.

7.17 Tubing for secondary containment shall be cleaned to ASTM G93 (Standard Practice for Cleaning Methods and Cleanliness Levels for Material and Equipment Used in Oxygen-Enriched Environments) level C to ensure no ignitable residue.

7.18 Material Certification

A certified copy of the material test report shall be provided at the time of shipment. The material test report shall include:

- Composition analysis (to include all elements required to be controlled by this standard)
- Grain size
- JK ratings of inclusion content
- Mechanical properties

7.19 Product Labelling

Material shall be identified with the following information:

- Manufacturer's name
- Purchaser's name and order number
- ASTM specification number(s)
- Heat number
- Material type (i.e., 316L)

7.20 Metal seal gaskets for VCR connections shall be made of electropolished stainless steel or nickel for all compatible applications. Gasket retainers shall be used when required by installation, generally in horizontal coupling, and site design construction procedures.

7.21 Carbon Monoxide (CO) is not compatible with nickel gaskets due to a reaction with nickel producing Nickel Penta-Carbonyl (process contaminant), and requires stainless steel VCR gaskets. For this reason, nickel, nickel-plated or "chrome-plated" gaskets should never be used for CO service. Similarly, for the same reason filter elements shall be made of stainless steel. All wetted components in CO service shall be made from stainless steel.

- 7.22 Nickel acts as a catalyst to decompose hydrides (AsH₃, B₂H₆, and PH₃). This reaction is more pronounced for low concentrations of the gas. For this reason, nickel or nickel-plated gaskets and filters should not be used for AsH₃, B₂H₆, PH₃, and any mixtures of these gases. Stainless steel (or compatible) gaskets and filters shall be used for these gases.
- 7.23 All metal wetted parts of UHP components (valves, pressure regulators, filter media, filter housings, pressure transducers, etc.) shall conform to the requirements as indicated in this design standard except for those applications where other materials are required for chemical compatibility and when requirements are clearly identified for specific materials. For example, diaphragms and poppets in regulators utilized in corrosive gases such as HBr, HCl, and others shall have Hastelloy C-22 trim, depending on vendor recommendations and engineering construction standards.
- 7.24 Manual valves shall be provided with an integral method for locking the valve in the closed position, and providing visual indication of OPEN or CLOSE position. Applicable to all manual valves within the gas supply system.
- 7.25 Process gas pneumatic valves shall be provided with a Rotowink pressure indicator to indicate the valve's OPEN or CLOSE position.
- 7.26 PFA ("Teflon") is the preferred material for pneumatic tubing for automatic valves. Applicable for all AOV valves within the gas supply systems.
- 7.27 For B₂H₆, HF, BF₃, Cl₂, ClF₃, HCl, HBr, DCS, 20%F₂/N₂, and SiCl₄, the "low pressure" valves (downstream of the pressure regulator) shall be the same model as the "high pressure valves (upstream of the pressure regulator). For these gases, all in-line process valves are to be the same.
- 7.28 Vent valves in high pressure gas service shall be the same model as the "high pressure" valves (upstream of the pressure regulator).
- 7.29 For Oxygen service
- a) Use valves with PTFE, PCTFE (Kel-F), or Vespel valve seats and Viton O-rings to reduce risks of oxygen fires.
 - b) Ensure all components are cleaned to ASTM G93 (Standard Practice for Cleaning Methods and Cleanliness Levels for Material and Equipment Used in Oxygen-Enriched Environments) level C to ensure no ignitable residue.
- 7.30 For valve and regulator seats in N₂O and CO₂ service, Vespel (Polyimide) is recommended because it swells to a lesser degree than Kel-F (PCTFE).

- 7.31 For valve and regulator seats in SiH₄ service, Vespel (Polyimide) is recommended. Kel-F (PCTFE) seats may swell and restrict flow. PCTFE seats may be used in gas cabinets due to their relatively low flow rates. Vespel seats are recommended for BSGS due to their higher flow rates.
- 7.32 For NH₃ service, Vespel (Polyimide) seats is not recommended.
- 7.33 Absolute pressure regulators shall be used for liquefied, low-pressure gases (i.e., WF₆, C₅F₈, SiCl₄, HF, ClF₃, DCS, etc.)
- 7.34 Regulators for BSGS shall use the "high flow" series model with a maximum outlet pressure $\geq$ 130 psig.
- 7.35 Recommended Manufacturers
 - a) Tubing Manufacturers: Valex, ~~Cardinal~~, Dockweiler, Swagelok, Sumikin, Kuze, Plymouth
 - b) Valve Manufacturers: Swagelok, AP Tech, Parker, Fujikin, Carten
 - c) Regulator Manufacturers: AP Tech, Tescom
 - d) Filter Manufacturers: Pall, Mott, Entegris, Nippon Seisen
- 7.36 Refer to "UHP Material List" for material recommendation by gas.

8.0 Welding

8.1 General Requirements

All welding should conform to the applicable requirements of the ASME Boiler and Pressure Vessel Code, Section IX, ANSI B16.25, B31.3 Chapter V, and AWS D10.9.

8.2 Testing of Welders

- a) All welds should be based upon Welding Procedure Specifications and be documented with associated Procedure Qualification Records in accordance with ASME Boiler and Pressure Vessel Code, Section IX, or with AWS D10.9.
- b) Qualification of the welding procedures to be used, and of the performance of the welders and welding operators, should conform to the requirements of the ASME Boiler and Pressure Vessel Code, Section IX, Articles II and III, or AWS D10.9.
- c) Welding and welder's certification shall be provided according to local codes and regulations, and site requirements.

8.3 Welding Procedure

a) Component Preparation

- i. All weld end preparation should be done in such a manner as to minimize the introduction of contaminants into the system. Make all necessary preparations for installing tubing and tube sub-assemblies so that exposure of the interior surface to atmosphere is minimized.
- ii. Weldment of assemblies are to be performed within a clean room environment. Materials are to be double bagged and wiped down when entering the clean room. Only open the bag within the clean room environment for component preparation and fabrication.
- iii. Use only tools and handling techniques that will not mar, disturb the shape of, or in any way reduce the quality of the materials used in this system.
- iv. All cutting of component or tubing weld ends should be done with a sharp-edged tool. No lubricants of any kind should be allowed. Cut ends shall be de-burred and faced to be square to an appropriate tolerance, per orbital welding equipment manufacturer's recommendation. The tubing shall not be scratched during de-burring. Abrasive wheels shall not be used. A lathe, milling, or portable end prep tool that clamps to the exterior of the tube is permitted.
- v. All component and tubing weld ends that are to be cut should be de-burred without chamfering.
- vi. All welding ends requiring preparation for procedures requiring the use of filler material should be prepared in accordance with ANSI/ASME B16.25.
- vii. All components should be maintained in a clean condition until welded into the system.

b) Welding Requirements

- i. Welding equipment used to make welds should be operated in accordance with the manufacturer's operating and safety instructions.
- ii. Welding shall be by the gas tungsten arc welding (GTAW) process with backing shielding and inside tubing purge. All stainless steel tubes shall be joined using an automatic orbital butt-welding machine.

c) Welding Procedures

- i. Prior to the welding of a particular size, wall thickness, alloy, and heat of material, sample welds should be made, dissected, and analyzed.
- ii. These welds should be checked for poor penetration, joint contamination, joint soundness, surface oxidation, discoloration, pitting, cracking, defects of fit-up, or defects of workmanship.
- iii. Once a sample weld is found to be acceptable, the sample weld should be the on-site work sample against which other welds of the same size, wall thickness, alloy, and heat of material are judged. All essential and supplementary essential variables should be documented in the procedure qualification record.
- iv. Any significant deviation(s) from the on-site work sample will cause the weld(s) to be rejected. Rejected welds should be removed and replaced.
- v. On-site work sample welds shall be made before each series of welding operations. On-site sample welds shall also be made at the start of each workday and any time any welding parameter is changed.
- vi. Sample test welds should be made periodically. These tests should be compared to the on-site work sample. Deviation from the on-site work sample should be cause for rejection. The sample test welds should be made at/when:
 - The beginning of each shift.
 - The end of each shift.
 - Any change in welding procedure, materials, tubing size, equipment, or equipment adjustments are made.
 - Any time that a concern or deviation to the weld quality as occurred.
- vii. Perform only one weld joint at a time.
- viii. Where a weld is found defective, the preceding two welds should be checked. If either of these welds shows signs of poor penetration, joint contamination, lack of joint soundness, surface oxidation, discoloration, pitting, cracking, defects of fit-up, or defects of workmanship, then all welds made since the last welding procedure was established should be removed and replaced.

d) Weld Set-up

Check parameters and verify that they are in accordance with the qualified welding procedure.

e) Purging

- i. A continuous nitrogen purge or other inert gas flow shall be maintained in piping system segments during cutting, facing, fabrication, and installation. Pulse purge shall be executed after cutting and facing to remove metal scraps and particles.
- ii. During welding, all tubes, fittings, valves, subassemblies, and all other components should be continuously purged. The entire piping system will be kept under a continuous purge until all welding is complete.
- iii. Automatic orbital head welding equipment should supply a constant gas-backing shield to the weld head during welding.
- iv. During all welding, a sufficient amount of purge/shield gas should be maintained until the weld head is removed from the newly welded parts.
- v. Argon for welding shall have 99.997% purity as a minimum with a moisture content of no more than 10.5 ppm.
- vi. Weld shield purge gas shall be filtered with a 0.003 μ filter to assure an inert environment
- vii. A backing shield (purge through the weld head) is required during welding.
- viii. Both purge/shield gas supply lines should contain flow indicators to ensure proper purging.
- ix. Tube ends shall be covered while the purge is removed using a technique that will minimize the amount of infiltration or contamination. Covers shall be of non-particulating material.

f) Weld Identification

- i. A daily log should be maintained on all welds, and as-built drawings recording all data should be maintained in the welding area.
- ii. All welds should be identified with a code number and cross-referenced with the drawings for future evaluation.

g) Additional Requirements

- i. Maintain sufficient distance between weld joints and valve seats to avoid damage to valve seats or valve stem tips.
- ii. Valves shall be located so as to allow space to operate the valve after installation at the job site.

9.0 Installation

9.1 Field Bending of Tubing

- a) Field bending of tubing is permitted for tube sizes up to 1 inch, with minimum 10x-tube diameter radius bend.
- b) Field bending of tubing is permitted for tube sizes of ¼ and 3/8 inch, with minimum 7.5x-tube diameter radius bend for inert and non-corrosive gases.
- c) Coaxial tubing for secondary containment must be bent so that the inner tube meets the above requirements.

9.2 Gasket retainers shall be used when required by installation conditions (i.e., in horizontal tubing runs) and site design construction procedures.

9.3 VCR locking systems shall be utilized to avoid radial torque on the VCR connection with potential leak, based on, but not limited to, the following situations:

- a) Where identified in the design documents.
- b) Any other specific application identified in the site construction specifications.

9.4 Gases incompatible with nickel (Ni) gaskets

Use stainless steel gaskets or other compatible material for the following gases and their mixes:

- a) CO
- b) AsH₃
- c) PH₃
- d) B₂H₆
- e) H₂Se

For a complete list of gases, refer to the UHP Materials List

9.5 Gases Requiring Secondary Containment

Refer to Global EHS – Toxic Gas Monitoring and Double Containment Standard, and Double Containment Matrix for list of spec gases that require secondary containment, and allowable exceptions.

9.6 Valves installed in non-HPM bulk gas distribution systems (N₂, O₂, Ar, He, CO₂) shall have downstream purge ports to allow for specific purging and venting during construction and qualification.

- 9.7 VCR connections are to be used as a coupling method in all bulk and specialty gas systems, and liquid chemical dispense and distribution systems such as TEOS and other precursors. VCR fittings shall be used when the sequence of installed components does not allow welding, when replacement of components is required for maintenance or upgrade, and when defined by system design and site construction specifications.
- 9.8 VCR locking systems shall be used to avoid radial torque on the VCR connections with potential for leak based upon, but not limited to, the following situations:
- a) Where clearly identified in the design documents.
 - b) Any other specific application identified in the site construction specifications.
- 9.9 Where feasible, bulk gas mains and laterals are to be pre-fabricated as pulled tee assembly including lateral POC valves for modular construction.

10.0 QA/Testing

10.1 Manufactured Equipment

- a) Stainless steel materials, components, and assemblies shall be manufactured and finally cleaned to meet high purity requirements. Manufacturers' procedures shall meet or exceed cleanliness based on, but not limited to, the following procedures:
 - i. Rinsed with 0.1 μ filtered, 18 megohm ultrapure water (UPW) and heated to 140°F (60°C) until the effluent meets manufacturer specification.
 - ii. Dried with 0.01 μ filtered nitrogen.
- b) Final assembly and packaging of materials and components shall be performed in an ISO Class 6 clean room space. Tubing shall be capped. Other components shall be appropriately capped or plugged depending on the port shape and configuration. Materials and components shall be bagged and purged with 0.01 μ filtered nitrogen or argon for shipment and storage.
- c) Welding of tubing and components shall be 100% tested by Micron, a contractor, or a third party qualified in Quality Control (QC). Testing, inspection, and acceptance criteria shall be identified in each site construction specification according to local codes and regulations.

- d) Weld coupons, weld tests, and programs shall be kept for tracking based on each QC program as defined in each site construction specification.
- e) Material certifications and specifications, machining and welding techniques, internal finishing process, and quality inspections/tests documents are to be maintained and provided.

10.2 Onsite Fabrication/Assembly

- a) Welding of tubing and components shall be 100% tested by Micron, a contractor, or third-party Quality Control (QC). Testing, inspection, and acceptance criteria shall be identified in each site construction specification according to local codes and regulations.
- b) Coupons, welding tests, and programs shall be kept for tracking based on each QC program as defined in each site construction specification.
- c) Leak tests and/or pressure tests, moisture tests, and particle tests shall be executed to guarantee integrity and quality of installations before releasing for process fluids.
- d) Proof pressure shall meet design pressure with 20% over capacity as a minimum.
- e) Moisture tests shall be performed for all new installations to ensure dry down as part of qualification tests. Maximum moisture concentration ≤ 40 ppb.
- f) Oxygen tests may not be routinely performed as it is assumed to follow the same decay trend of moisture. Oxygen test requirements (if needed) shall be identified in the site construction specifications and directed during the design phase.
- g) Particle contamination shall be controlled to ensure that pipelines, manifolds, and assemblies have been appropriately built, installed, and pre-cleaned. A shock purge procedure shall be performed to increase particle removal efficiency as an auxiliary procedure for final qualification and shall be defined in each site construction specification.
- h) Particle measurement shall be executed based on, but not limited to, the following methods:
 - i. Steady flow measurements
 - Laser Particle Counter (LPC)
 - Analyze 1 ft³ per run. Three consecutive runs, as a minimum. It is recommended to utilize particle counters running at 1 ft³/min to minimize qualification time.
 - ≤ 3 counts of 0.2 μ and larger particles.

- Condensation Nucleus Counter (CNC)
 - Analyze 1 ft³ per run. Three consecutive runs, as a minimum.
 - ≤ 3 counts of 0.01 μ and larger particles.
- ii. Pulse Purge Measurement
 - Laser Particle Counter (LPC)
 - ≤ 10 counts of 0.2 μ and larger particles for 5 consecutive 1-minute sampling periods.
- i) Leak tests shall be executed according to SEMI F1 and local codes and regulations (i.e., ASME B31.3 345.1) as follows:
 - i. Inboard helium leak check: Pull high vacuum into the line to be tested, spray helium close and across welded and mechanical connections. Leak rates not to exceed 1×10^{-9} Atm cc/sec.
 - ii. Outboard leak check: Fill the line to be tested with a helium or helium mixture. Sniff each weld and mechanical connection. Readings above 1×10^{-6} Atm cc/sec should be considered a leak.
- j) If pressure testing is required, it shall be performed as directed by site construction specifications.

11.0 Fluorine Passivation

11.1 General

The following process gases require fluorine passivation of the tubing and UHP components used to contain and convey these gases:

- a) 20%F₂/N₂ (or greater concentration)
- b) F₂/Kr/Ne
- c) F₂/Ar/Ne
- d) HF
- e) ClF₃
- f) WF₆
- g) BF₃

For a complete list of gases, refer to the UHP Materials List (**Appendix 2**)

11.2 Fluorine Passivation Procedure Example

- a) Site Gas Team and contractor shall collaborate and agree on Fluorine Passivation Procedure to be used. Consult equipment manufacturer for recommended passivation procedure for their equipment. Procedure shall be specific for the gas supply system (cabinet or BSGS) and VMB/DVB equipment operations. Consideration shall be given to the proper disposal and/or abatement of the passivation gas during evacuation.
- b) Fluorine passivation shall include the gas cabinet, VMB, and distribution tubing between the gas cabinet to VMB and between the VMB to the tool (or tool's GIB).
- c) Procedure shall include the following steps:
 - i. Perform cycle purges (pressurization/vacuum) and leave the equipment/line under vacuum.
 - ii. Pressurize with passivation gas (typically a 1% to 5% F2 mixture)
 - iii. Hold pressure and allow gas to "soak" for a specified duration.
 - iv. Evacuate the passivated gas and leave the equipment/line under vacuum.
 - v. Repeat the above process of pressurization, soak, and evacuation. This should be repeated for multiple cycles. For each subsequent cycle, the "soak" period should increase up to 4 hours in duration.
 - vi. For thorough passivation, the total elapsed "soak" time for the equipment/line should be > 60 hours.
 - vii. For long gas lines (> 25 m), add another 24 hours (with 4 hour "soak" durations) for every 25 m.

12.0 Process or Tool Specific Requirements

12.1 HF Gas Supply for 21L Vapor Etch (TEL LEAGA) Applications

Refer [to Global Facilities – O&M BKM – Gas – HF Gas System Guidelines for 21L Vapor Etch Improvement](#) for design requirements specific to this tool.

13.0 Document Control

13.1 Approval: GLOBAL_FAC_GFTT_APPR

13.2 Notification:

GLOBAL_FAC_MANAGERS

GLOBAL_FAC_COMMUNICATIONS

GLOBAL_FAC_CONSTRUCTION

GLOBAL_FAC_NOTIFY

FCG_F4_FAC_NOTIFY

FCG_F6_FAC_NOTIFY

FCG_F10_FAC_NOTIFY

FCG_F11_FAC_NOTIFY

FCG_F15_FAC_NOTIFY

FCG_F16_FAC_NOTIFY

FCG_MMP_FAC_NOTIFY

FCG_MMY_FAC_NOTIFY

FCG_MSB_FAC_NOTIFY

FCG_MXA_FAC_NOTIFY

FCG_MTB_FAC_NOTIFY

13.3 Final Viewer Access

All Micron Team Members at all Sites

13.4 Retention

Adherence to the requirements specified in

[Global Facilities - Document Control Procedures](#)

13.5 Review

Biennially (two years)

14.0 Revision History

Revision #	Section Changed	Details of Changes	Date
0	NA	New document	09/12/22

15.0 APPENDIX

APPENDIX 1

Template Form for Site Specific Deviation List

Site: _____

No	Specification Section and Item	Deviation Proposal	Scope Impact (Benefit, Cost, Schedule)	Remark	Approval: 1. Site 2. GFTT 3. FCT

APPENDIX 2

- UHP Material List: [LINK](#)